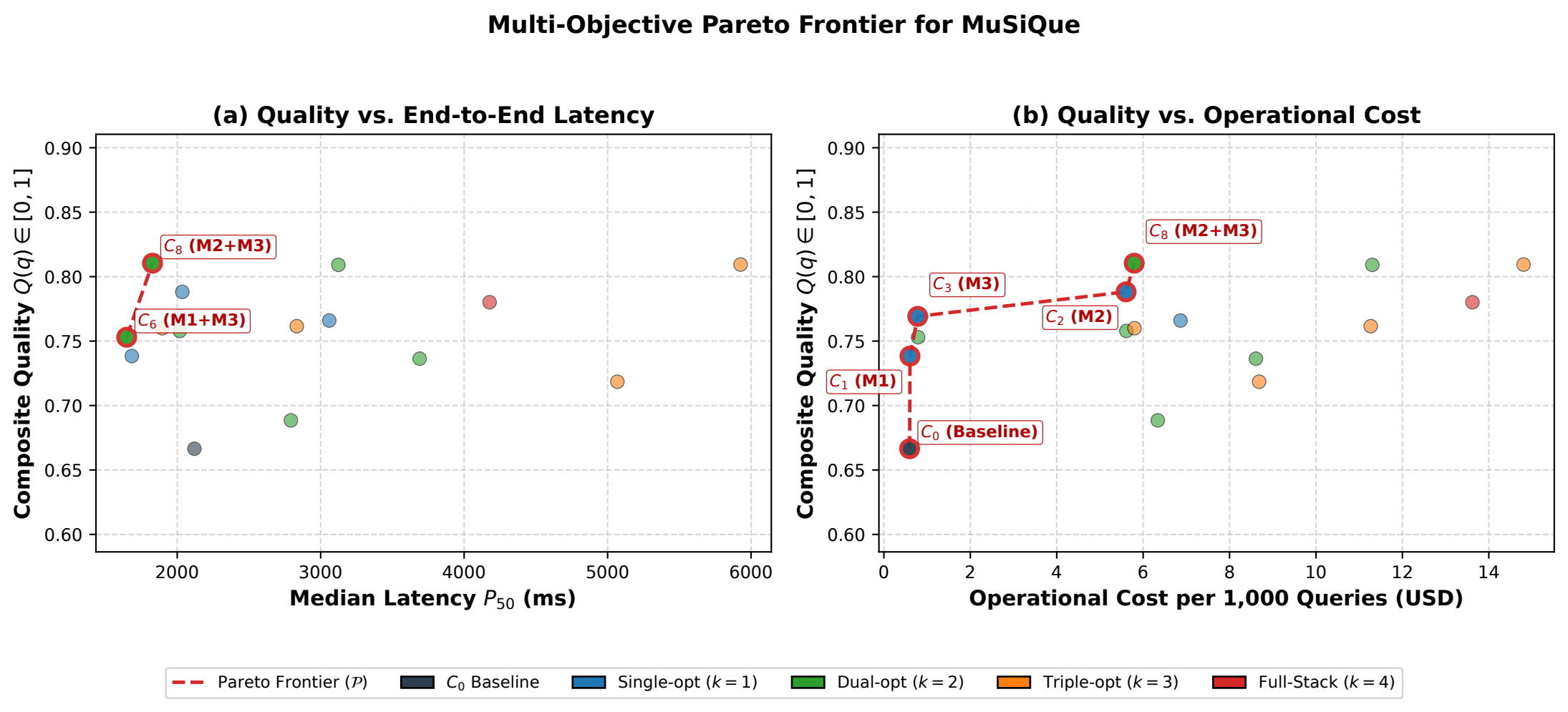
Multi-Objective Pareto Frontier for MuSiQue
(a) Quality vs. End-to-End Latency
(b) Quality vs. Operational Cost
Composite Quality $Q(q) \in [0, 1]$
Median Latency $P_{50}$ (ms)
Operational Cost per 1,000 Queries (USD)
$C_8$ (M2+M3)
$C_6$ (M1+M3)
$C_3$ (M3)
$C_2$ (M2)
$C_1$ (M1)
$C_0$ (Baseline)
Pareto Frontier ($\mathcal{P}$)
$C_0$ Baseline
Single-opt ($k = 1$)
Dual-opt ($k = 2$)
Triple-opt ($k = 3$)
Full-Stack ($k = 4$)